



Maritime &
Coastguard
Agency

Guidance

Annex A: Requirements from SOLAS Chapter VI

Updated 28 May 2024

Contents

1. Cargo information
2. Verified Gross Mass (VGM)
3. Stowage and Securing – Packing of Units and Cargo Securing Manual
4. Solid Bulk Cargoes - Stability information and Cargo loading manual
5. Solid Bulk Cargoes - International Maritime Solid Bulk Cargoes (IMSBC) Code
6. Solid Bulk Cargoes - International Code for the Safe Carriage of Grain in Bulk
7. Solid Bulk Cargoes - Code of Practice for Safe Loading and Unloading of Bulk Carriers (BLU Code)
8. Code of Safe Practice for Cargo Stowage and Securing (CSS Code), Code of Safe Practice for Ships Carrying Timber Deck Cargoes (TDC Code) and other IMO publications
9. Exemptions and Equivalents - Vessels
10. Equipment approval – Gas Detectors
11. Test Facility Approval – Testing related to the Carriage of Cargoes

12. Material Safety Data Sheets - Oil
13. Prohibition on blending of bulk liquid cargoes
14. The safe use of pesticides on ships



© Crown copyright 2024

This publication is licensed under the terms of the Open Government Licence v3.0 except where otherwise stated. To view this licence, visit nationalarchives.gov.uk/doc/open-government-licence/version/3 or write to the Information Policy Team, The National Archives, Kew, London TW9 4DU, or email: psi@nationalarchives.gov.uk.

Where we have identified any third party copyright information you will need to obtain permission from the copyright holders concerned.

This publication is available at <https://www.gov.uk/government/publications/mgn-699-m-guidance-on-the-merchant-shipping-carriage-of-cargoes-regulations-2024/annex-a-requirements-from-solas-chapter-vi>

1. Cargo information

1.1 SOLAS Chapter VI, regulation 2 sets out a fundamental requirement in the carriage of cargoes, namely that a shipper or forwarder must provide the master or their representative with appropriate information sufficiently in advance of loading to enable a proper assessment of the suitability of the cargo for the ship on its intended voyage including stowage and segregation considerations.

1.2 The Merchant Shipping (Carriage of Cargoes) Regulations 2024 and the Merchant Shipping (Dangerous Goods and Harmful Substances) (Amendment) Regulations 2024, taken together, regulate the minimum information and documentation which must be provided by shippers and freight forwarders.

1.3 The International Maritime Organization (IMO) has published generic model forms to assist shippers in the presentation of cargo information, such as the form for Cargo Information for Solid Bulk Cargoes found in section 4.2.3 of the IMSBC Code (the information is mandatory under the code, but the form set out in the template is recommendatory). Stakeholders should be aware, however, that a cargo information form is not suitable for packaged dangerous goods requiring a dangerous goods

declaration in accordance with SOLAS Chapter VII. An example of a multimodal dangerous goods form can be found in The International Maritime Dangerous Goods (IMDG) Code at section 5.4.5 and a template can also be found on www.gov.uk (<https://www.gov.uk/government/publications/dangerous-goods-declaration-note>).

1.4 As an aid to digitization of shipping documentation, SOLAS Chapter VI also provides that Electronic Data Processing (EDP) or Electronic Data Interchange (EDI) methods may be used by shippers for providing the cargo information.

2. Verified Gross Mass (VGM)

2.1 SOLAS Chapter VI, regulation 2 requires that the gross mass of cargo (contents) carried in units, including tare weight of the container, is verified. Marine Guidance Note, MGN 534, gives detailed advice on how UK shippers, port and terminal operators, and carriers can meet their VGM responsibilities calibration services.

2.2 In this section of SOLAS Chapter VI, in order to clarify the meaning of the word “container”, reference is made to another IMO Convention, namely the International Convention for Safe Containers (CSC), 1972. In the UK, the Health and Safety Executive (HSE) oversees the implementation of CSC Convention requirements such as freight container design, marking, plating and maintenance.^[footnote 1]

3. Stowage and Securing – Packing of Units and Cargo Securing Manual

3.1 SOLAS Chapter VI, regulation 5 requires that cargoes are packed and secured within cargo units and cargo transport units in such a way as to prevent, throughout the voyage, damage or hazard to the ship and the persons on board. This is a key responsibility for parties ashore undertaking cargo handling and consigning activities, especially in relation to cargo transport units involved in multimodal transportation that includes a sea carriage leg.

3.2 SOLAS Chapter VI, regulation 5 also requires that all cargoes, cargo units and cargo transport units are loaded, stowed and secured throughout the voyage in accordance with the ship's Cargo Securing

Manual which must be approved by the Administration. Approval of the Cargo Securing Manual for UK ships is carried out by the Maritime and Coastguard Agency (MCA) for unclassified vessels and, where they are the Classification Society, the UK Recognised Organisations are authorised by the MCA to undertake this approval for classed vessels. Details on the UK Recognised Organisations can be found on the MCA's website.^[footnote 2]

3.3 When approval by the MCA is required, a ship operator should submit the ship's Cargo Securing Manual to the Marine Office responsible for the survey of a new building or nearest to the port from which the ship operates. Fees are chargeable for approval based on the published hourly rate.^[footnote 3]

3.4 Annex C (<https://www.gov.uk/government/publications/mgn-699-m-guidance-on-the-merchant-shipping-carriage-of-cargoes-regulations-2024/annex-c-stowage-and-securing-cargo-securing-manual>) to this MGN contains further guidance including advice for stakeholders on the structure and layout of a Cargo Securing Manual and indicates some acceptable alternatives and modifications. Annex D (https://assets.publishing.service.gov.uk/media/664f23b0b7249a4c6e9d3a3e/MGN_699_Annex_D.pdf) to this MGN contains a general exemption for cargo ships below 500GT not engaged on international voyages from the requirements for a Cargo Securing Manual specified in SOLAS Chapter VI, regulation 5.6. This exemption does not extend to any other stowage and securing requirements specified in SOLAS Chapter VI, regulation 5, nor does it extend to the requirements for a Cargo Securing Manual specified in other SOLAS chapters and safety regulations such as those concerning the carriage of dangerous goods.

4. Solid Bulk Cargoes - Stability information and Cargo loading manual

4.1 SOLAS Chapter VI, regulation 6 requires the owner to ensure that the master is provided with comprehensive information on ship's stability and cargo distribution prior to loading a solid bulk cargo. Furthermore, reflecting the provision in SOLAS Chapter VI, regulation 7.2, there is a requirement to ensure that the ship is provided with a cargo loading manual, written in a language with which the ship's officers responsible for cargo operations are familiar. If that language is not English, the ship must also be provided with a manual written in the English language. The manual may consist of one or more booklets and should, as a minimum, include:

- a) stability data, to the extent required by SOLAS Chapter II-1 regulation 5-1 'Stability Information to be Supplied to the Master' (see regulation 32 of the Merchant Shipping (Load Line) Regulations 1998).
- b) ballasting and deballasting rates and capacities;
- c) maximum allowable load per unit surface area of the tank top plating;
- d) maximum allowable load per hold;
- e) general loading and unloading instructions with regard to the strength of the ship's structure including any limitations on the most adverse operating conditions during loading, unloading, ballasting operations and the voyage;
- f) any special restrictions such as limitations on the most adverse operating conditions imposed by the Administration or organization recognised by it, if applicable; and
- g) where strength calculations are required, maximum permissible forces and moments on the ship's hull during loading, unloading and the voyage.

4.2 Approval of vessel stability information is undertaken by the Maritime and Coastguard Agency's stability unit. Guidance for owners and naval architects on how and where to submit ship stability information for MCA authorisation can be found on www.gov.uk (<https://www.gov.uk/guidance/approval-of-vessel-stability-information-mca-stability-unit>).

5. Solid Bulk Cargoes - International Maritime Solid Bulk Cargoes (IMSBC) Code

5.1 SOLAS Chapter VI, regulation 1-2 requires that the carriage of solid bulk cargoes other than grain must comply with the relevant provisions of the International Maritime Solid Bulk Cargoes Code ("the IMSBC Code"). The IMSBC Code became mandatory on an international basis and replaced the previous Code of Safe Practice for Solid Bulk Cargoes ("BC Code") when it entered into force on 1 January 2011.

5.2 The IMSBC Code is reviewed and updated biennially through the International Maritime Organization ("IMO"). Amendments are adopted by means of Maritime Safety Committee ("MSC") Resolutions which set out

both the textual changes and associated dates. In the case of IMSBC Code amendments, the MSC Resolutions customarily state that they may be applied by Governments on a voluntary basis from 12 months before the entry into force date. Accordingly, the MCA supports the voluntary early uptake of amendments to the IMSBC Code by industry, especially when considering the wider practicalities of training and familiarisation with the latest published editions of the Code.

5.3 Since 2011, the MCA has issued various Merchant Shipping Notices to advise and guide stakeholders of pertinent changes to the IMSBC Code and will continue to do so as required from time to time. Stakeholders should familiarise themselves with these documents as appropriate.

5.4 Solid bulk cargoes should only be shipped in accordance with the IMSBC Code and only those solid bulk cargoes with schedules in the Code may be shipped.

5.5 Where a solid bulk cargo is not listed in the Code but its shipment in bulk is required, section 1.3 of the IMSBC Code applies. For solid bulk cargoes not listed in the Code, all details, characteristics, test results, classifications etc. on the product should be provided to the competent authority of the port of loading. The competent authority of the port of loading will use this data to identify appropriate carriage requirements in line with the general provisions of the IMSBC Code.

5.6 Where the product is classified as Group A, Group B or Group A and B, the competent authority of the port of loading should contact the flag state of the vessel(s) to be used for this cargo and the competent authority of the port(s) of unloading to request their agreement and, where appropriate, any additional conditions or carriage requirements. For cargoes that have been classified as Group C, the competent authority of the port of loading should notify the flag state(s) and the competent authority of the port(s) of unloading of the product, its assessment as a Group C cargo and carriage requirements.

5.7 For unlisted solid bulk cargoes where the port of loading is in the United Kingdom, shippers should contact Cargoes@mcga.gov.uk in the first instance and should refer to MGN 512^[footnote 4] for details of the information to be provided.

6. Solid Bulk Cargoes - International Code for the Safe Carriage of Grain in Bulk

6.1 Regulations 8 and 9 (Part C – Carriage of grain) of SOLAS Chapter VI require cargo ships carrying grain to comply with the requirements of the International Code for the Safe Carriage of Grain in Bulk (“the International Grain Code”) and to hold a Document of Authorization as required by that Code. The International Grain Code was adopted by the IMO through Resolution MSC.23(59) and is implemented in the UK by the Merchant Shipping (Carriage of Cargoes) Regulations 2024.

6.2 The term ‘grain’ includes wheat, maize (corn), oats, rye, barley, rice, pulses, seeds and processed forms thereof whose behaviour is similar to that of grain in its natural state. A ship carrying grain in bulk must have available on board a document of authorization in compliance with the International Grain Code, an approved Grain Stability Booklet/a grain loading manual and a copy of the International Grain Code.

6.3 The Alternative Compliance Scheme (ACS) or the Enhanced Authorisation Scheme (EAS) may apply to UK registered vessels which trade internationally and which meet the conditions set out in MGN 568(M) ^[footnote 5] and MGN 561(M) ^[footnote 6] respectively. Under the ACS or EAS, the Document of Authorization for the Carriage of Grain may be issued by one of the UK’s authorised Recognised Organisations. Details on the UK’s Recognised Organisations can be found on the MCA’s website. ^[footnote 7] The UK’s Recognised Organisations may also approve Grain Stability Booklets.

6.4 SOLAS Chapter VI, regulation 9.2 prohibits a ship without a Document of Authorization for the carriage of grain from loading grain unless the master satisfies the Administration that the ship will comply with the requirements of the International Grain Code in its proposed loaded condition. Where a UK registered ship is relying on this provision in order to carry grain in bulk, the master should, in the first instance, contact the relevant MCA Marine Office with a document signed by a surveyor of the certifying authority stating that the ship in its proposed loaded condition will comply with the requirements of the International Grain Code. Only if an MCA surveyor is satisfied that the appropriate detailed requirements have been met will such an approval be given.

7. Solid Bulk Cargoes - Code of Practice for Safe Loading and Unloading of Bulk Carriers (BLU Code)

7.1 SOLAS Chapter VI, regulation 7 addresses the loading, unloading and stowage of solid bulk cargoes including grain and references the IMO's "Code of Practice for Safe Loading and Unloading of Bulk Carriers" as amended ("BLU" Code adopted by Resolution A.862(20)) which contains detailed operational guidance, including the responsibilities of the master and the terminal representative in the cargo operations.

7.2 Further details including the requirements for agreeing loading and unloading plans can be found in Annex B

(<https://www.gov.uk/government/publications/mgn-699-m-guidance-on-the-merchant-shipping-carriage-of-cargoes-regulations-2024/annex-b-loading-unloading-and-stowage-of-solid-bulk-cargoes-and-the-blu-code>) to this MGN.

8. Code of Safe Practice for Cargo Stowage and Securing (CSS Code), Code of Safe Practice for Ships Carrying Timber Deck Cargoes (TDC Code) and other IMO publications

8.1 SOLAS Chapter VI, regulation 1 refers to both the IMO's Code of Safe Practice for Cargo Stowage and Securing, as amended ("CSS Code" adopted by Resolution A.714(17)) and the IMO's Code of Safe Practice for Ships Carrying Timber Deck Cargoes ("TDC Code" adopted by Resolution A.1048(27)).

8.2 As a general principle, the owner and master of every ship to which SOLAS Chapter VI applies must ensure that appropriate documentation relevant to the carriage, stowage and securing of cargo is carried on board. Such documentation will specify the precautions necessary for the safe carriage of that cargo by sea.

8.3 As such, for all vessels subject to and complying with the requirements of SOLAS Chapter VI, the MCA expects copies of both relevant and up to date guidance to be available on board such as the IMO's cargo related codes including but not limited to the CSS Code, the TDC Code, the IMSBC Code, the International Grain Code, the Code of Safe Practice for the Carriage of Cargoes and Persons by Offshore Supply Vessels ("OSV Code" adopted by Resolution A.863(20)) and the IMO/ILO/UNECE Code of Practice for Packing of Cargo Transport Units ("CTU Code" published by the IMO as MSC.1/Circ.1497).

8.4 The provision and use of such relevant cargo guidance may also be beneficial when developing plans for key shipboard operations concerning the safety of the ship and the prevention of pollution required under SOLAS Chapter IX and the International Safety Management (ISM) Code.

9. Exemptions and Equivalents - Vessels

9.1 SOLAS Chapter VI, regulation 1 provides that for cargo ships of less than 500 gross tonnage an Administration, where it considers that the sheltered nature and conditions of voyage would make application of any specific requirements of part A (General Provisions) or part B (Special Provisions for Solid Bulk Cargoes) of SOLAS Chapter VI unreasonable or unnecessary, may take other effective measures to ensure safety.

9.2 Reflecting this SOLAS provision and in keeping with other merchant shipping safety legislation, the 2024 Regulations contain provisions which enable the Secretary of State to grant exemptions or approve equivalents in certain circumstances. Applications for an exemption or equivalence can be made in the usual way using Maritime and Coastguard Agency form MSF 1261, which may be obtained from the nearest MCA Marine Office (<https://www.gov.uk/government/organisations/maritime-and-coastguard-agency/about/access-and-opening#marine-offices><https://www.gov.uk/government/publications/locations-of-mca-marine-offices>). An exemption may be granted subject to such safety requirements as the MCA considers necessary to ensure the overall safety of the vessel. For equivalences, any fitting, material, appliance, apparatus, arrangement or other provision is required to be at least as effective as that required by SOLAS Chapter VI.

10. Equipment approval – Gas Detectors

10.1 SOLAS VI, regulation 3 requires the provision of instruments for oxygen analysis and gas detection when carrying bulk cargoes liable to emit toxic or flammable gases or cause oxygen depletion in the cargo hold. The crew of a ship should be familiar with the use of the instruments and such instruments should be regularly used during cargo monitoring. The instruments should be regularly serviced and calibrated in accordance with the manufacturers' instructions.

10.2 SOLAS VI, regulation 3 requires that such an instrument must be to the satisfaction of the Administration. For information on design, construction and performance requirements, and testing standards, refer to MSN 1874.^[footnote 8] For information on approved equipment, please refer to the UK's marine equipment database.

(<https://www.gov.uk/government/publications/uk-marine-equipment-approval-database>).

11. Test Facility Approval – Testing related to the Carriage of Cargoes

11.1 For test facilities in the UK, the United Kingdom Accreditation Service (“UKAS”) is the sole national accreditation body recognised by the UK Government to assess the competence of organisations that provide certification, testing, inspection and calibration services.

11.2 For further information on UK accredited organisations providing cargo testing services, please refer to the UKAS website (<https://www.ukas.com/find-an-organisation/>).

11.3 Where organisations providing cargo testing services are located in the UK but are not UKAS accredited, other relevant industry standards or accreditations may be acceptable, subject to their being agreed with the MCA prior to any test documentation being relied upon in relation to the shipment of a cargo by sea. Enquiries of this nature should be directed to Cargoes@mcga.gov.uk.

11.4 Otherwise, the results of verification and tests carried out by bodies or laboratories of other IMO member States offering suitable and satisfactory guarantees of technical and professional competence and independence will be accepted.

12. Material Safety Data Sheets - Oil

12.1 SOLAS Chapter VI, regulation 5-1 requires ships carrying MARPOL Annex I products, such as marine fuel oils, to be provided with a material safety data sheet prior to the loading of such cargoes or fuel oils. The material safety data sheet must be provided to the ship prior to the loading of such oil as cargo in bulk or the bunkering of oil fuel.

12.2 The format of the material safety data sheet can be found in the IMO Recommendations for Material Safety Data Sheets (MSDS) for MARPOL Annex I Oil Cargo and Oil Fuel (MSC.286(86)).

13. Prohibition on blending of bulk liquid cargoes

13.1 SOLAS Chapter VI, regulation 5-2 prohibits the physical blending and production processing of bulk liquid cargoes on board a ship during sea voyages.

13.2 Physical blending refers to the process whereby the ship's cargo pumps and pipelines are used to internally circulate two or more different cargoes with the intent to achieve a cargo with a new product designation. Production processes refer to any deliberate operation whereby a chemical reaction between a ship's cargo and any other substance or cargo takes place.

13.3 SOLAS Chapter VI, regulation 5-2 also provides for exceptions to the prohibition, namely that it does not apply to the blending of products or the production processes of cargoes for use in the search and exploitation of seabed mineral resources on board ships used to facilitate such operations. A well stimulation vessel would be an example of such a ship. In keeping with the general principles of the SOLAS Convention, it is also stated that the prohibition does not preclude the master from undertaking cargo transfers for the safety of the ship or protection of the marine environment.

13.4 The IMO Circular 'Guidelines for the Carriage of Blends of Biofuels and MARPOL Annex I Cargoes' (MSC-MEPC.2/Circ.17) draws attention to this SOLAS Chapter VI prohibition and further states that such blending operations should only be undertaken whilst a ship is within port limits.

13.5 Within port limits, such operations are subject to any local port regulations and bye-laws. Stakeholders are strongly advised to check ahead with the relevant authorities when planning such cargo operations.

14. The safe use of pesticides on ships

14.1 SOLAS VI, regulation 4 requires that appropriate precautions are taken in the use of pesticides in ships and reference is made to the following three IMO publications:

- Revised Recommendations on the safe use of pesticides in ships (MSC.1/Circ.1358);
- Recommendations on the safe use of pesticides in ships applicable to the fumigation of cargo holds (MSC.1/Circ.1264, as amended by MSC.1/Circ.1396); and
- Revised Recommendations on the safe use of pesticides in ships applicable to the fumigation of cargo transport units (MSC.1/Circ.1361/Rev.1).

14.2 Accordingly, where pesticides are used in the cargo spaces of ships prior to, during or following a voyage, the 2024 Regulations and MSN 1917 require that precautions are established with regard to the relevant IMO Recommendations. As per the IMO's Maritime Safety Committee, these Recommendations are intended to supplement the requirements of the IMDG code, the IMSBC code and the International Grain Code.

14.3 Given the breadth of this topic and its associated safety concerns, further detailed guidance is contained in our separate Marine Guidance Note MGN 700.

-
1. <https://www.hse.gov.uk/ports/regulations.htm>
(<https://www.hse.gov.uk/ports/regulations.htm>)
 2. [List of UK Recognised Organisations \(https://www.gov.uk/guidance/uk-recognised-organisations\)](https://www.gov.uk/guidance/uk-recognised-organisations)
 3. <https://www.gov.uk/government/publications/maritime-coastguard-agency-mca-fees> (<https://www.gov.uk/government/publications/maritime-coastguard-agency-mca-fees>)
 4. [MGN 512 – Solid Bulk Cargoes: Cargoes not Listed in the International Maritime Solid Bulk Cargoes \(IMSBC\) Code](https://www.gov.uk/government/publications/mgn-512m-solid-bulk-cargoes)
(<https://www.gov.uk/government/publications/mgn-512m-solid-bulk-cargoes>)
 5. [MGN 568\(M\) - Surveys - Alternative Compliance Scheme](https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/877711/MGN_568_ACS_-_Final_March_2017.pdf)
(https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/877711/MGN_568_ACS_-_Final_March_2017.pdf)
 6. [MGN 561\(M\) - Surveys - Enhanced Authorisation Scheme](https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/604839/MGN561_Enhanced_Authorisation_-_Final_March_2017.pdf)
(https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/604839/MGN561_Enhanced_Authorisation_-_Final_March_2017.pdf)

7. List of UK Recognised Organisations ([https://www.gov.uk/guidance/uk-
authorised-recognised-organisations-ros](https://www.gov.uk/guidance/uk-
authorised-recognised-organisations-ros))
8. MSN 1874 (M&F) Amendment 8
([https://www.gov.uk/government/publications/msn-1874-mf-amendment-8-
marine-equipment-united-kingdom-conformity-assessment-procedures-for-
marine-equipment-other-approval-and-standards](https://www.gov.uk/government/publications/msn-1874-mf-amendment-8-
marine-equipment-united-kingdom-conformity-assessment-procedures-for-
marine-equipment-other-approval-and-standards)) – Marine Equipment -
United Kingdom Conformity Assessment Procedures for Marine
Equipment – Other Approval Standards



OGI

All content is available under the [Open Government
Licence v3.0](#), except where otherwise stated



© Crown copyright